

Week 1: Logic

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- Propositions, truth values
- Negations (not)
- Conjunction (and), disjunction (or)
- Conditional statement (if ... then ...) and converse
- Bi-conditional statement
- De Morgan's Law

1.1. Propositional Logic

Definition

A **proposition** is a declarative sentence that is either true or false, **but not both**.

Example. The following are propositions.

- Kuala Lumpur is the capital of Malaysia.
- New York is the capital of the United States of America.
- $1 + 1 = 2$.
- $2 + 2 = 3$. **F**

Example. The following are not propositions.

- What time is it? This is a question. It has no truth value.
- Read this carefully. We cannot tell whether it is T or F.

This is an instruction. Again it has no truth value.

Example. The following are not propositions.

- What time is it?
- Read this carefully.
- $x + 1 = 2$. ($x+1 = 2$) is not a proposition; this is an equation.
- $x + y = z$. Here x, y, z are variables ranging in real numbers.
values can be changed

When $x = 1$, then $x + 1 = 2$, so the statement is T.
When x is not equal to 1, then $x + 1$ is not equal to 2,
so the statement is F.

The same applies to $(x + y = z)$; this is a multi-variable equation.

We usually use letters $p, q, r, s, u, \dots$ to represent propositions.

The **truth value** of a proposition is T if it is true, and F if it is false.

$$TV : \{p, q, r, s, \dots\} \rightarrow \{T, F\}.$$

Example. p : Kuala Lumpur is the capital of Malaysia.

$$TV(p) = T, \text{ or we say: } p \text{ is true.}$$

for short: $p = T$

Example. q : Kuala Lumpur is the capital of Thailand.

$$TV(q) = F, \text{ or we say: } q \text{ is false.}$$

for short: $q = F$

Definition

Let p be a proposition. The **negation** of p , denoted by $\neg p$, is read as “not p ”. verb: negate $\sim p$

Example. p : Mei has an MP3 player.

$\neg p$:

Definition

Let p be a proposition. The **negation** of p , denoted by $\neg p$, is read as “not p ”.

Example. p : Mei has an MP3 player.

$\neg p$: Mei does not have an MP3 player.

Example. q : Steve has more than 100 GB free disk space on his laptop. Steve has less than 100 GB (not correct)

$\neg q$: Steve does not have more than 100 GB
Steve has less than or equal to 100GB
Steve has at most 100 GB.

$\sim(\dots \text{less than } \dots) = \dots \text{at least } \dots$

Example. r : It is below freezing and snowing.

$\neg r$:

negate both

It is not below freezing or not snowing.

‘and’ changed to ‘or’

These are not correct:

It is not below freezing and snowing.

It is not below freezing or snowing.

Example. r : It is below freezing and snowing.

$\neg r$:

Example. s : The prerequisite for the master course is Number Theory or Cryptography.

$\neg s$:

The prerequisite is not number theory and not Cryptography

English: The prerequisite ... is neither Number Theory nor Cryptography.

Example. t : I come to the university every day.

$\neg t$:

I do not come to the university for at least one day

I do not come to the university for some day

There exists (at least) one day, I do not come to the university.

Truth Table

p	$\neg p$
T	F
F	T

Truth Table

	p	$\neg p$	$\sim\sim p = \sim(\sim p)$
Case 1:	T	F	T
Case 2:	F	T	F

The first and last columns have the same truth values in both cases.

Question: What is $\neg(\neg p)$?

$$\sim\sim p = p$$

Binary connectives

Definition

Let p and q be propositions. The **conjunction** of p and q is the proposition “ p and q ”, denoted by $p \wedge q$.

And

Binary connectives

Definition

Let p and q be propositions. The **conjunction** of p and q is the proposition “ p and q ”, denoted by $p \wedge q$.

Example Let

p : You have the flu.

q : You miss the concert.

Find the conjunction of p and q .

(p and q) : You have the flu and you miss the concert.

Example: p : It is below freezing; q : It is snowing.

(p and q): It is below freezing and snowing.

Example Let

$$p : x \geq 10, \quad q : x < 20.$$

$$r : x \text{ is an integer.}$$

$$p \wedge q: \quad x \geq 10 \text{ and } x < 20.$$

You may also say: $20 > x \geq 10.$ $10 \leq x < 20.$

Example Let

$$p : x \geq 10, \quad q : x < 20.$$

$$r : x \text{ is an integer.}$$

$$p \wedge q:$$

$$\begin{aligned} p \wedge q \wedge r: \quad & p \text{ and } q \text{ and } r = (p \text{ and } q) \text{ and } r \\ & = (20 > x \geq 10) \text{ and } x \text{ is an integer.} \end{aligned}$$

Another way to say:

$$p \text{ and } q \text{ and } r = (x \text{ is equal to } 10, 11, 12, 13, 14, 15, 16, 17, 18, 19)$$

Example Let

$$p : x \geq 10, \quad q : x < 20.$$

$$r : x \text{ is an integer.}$$

$$p \wedge q:$$

$$p \wedge q \wedge r:$$

Example Let

$$p : x \text{ is a real number.}$$

$$q : x^2 = -1.$$

$$p \wedge q: \quad x \text{ is a real number and } x^2 = -1.$$

(p and q): x does not exist.

(p and q) is a false statement, $(p \text{ and } q) = F$.

Definition

Let p and q be propositions. The **disjunction** of p and q is the proposition “ p or q ”, denoted by $p \vee q$.

Note: Disjunction is the *inclusive or*, which means that both statements can be true.

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Note: Disjunction is the *inclusive or*, which means that both statements can be true.

Example Let

p : You have the flu.

q : You pass the course.

	flu	pass
(p or q) covers 3 cases	O	O
	O	X
	X	O
	X	X

Find the disjunction of p and q .

(p or q): You have the flu or you pass the course.

Example

You eat; otherwise, you die.

p

or

q

(p or q) covers three cases.

eat

die

O

O

(possible)

O

X

(most likely)

X

O

(reasonable consequence)

p or q does not cover:

X

X

(impossible)

In contrast, there is also the *exclusive or*: exactly one of the statements is true.

Definition

Let p and q be propositions. The **exclusive or** of p and q is the proposition “ p XOR q ”, denoted by “ $p \oplus q$ ”.

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Definition

Let p and q be propositions. The **exclusive or** of p and q is the proposition “ p XOR q ”, denoted by “ $p \oplus q$ ”.

Example. Menu: Soup or Salad.

p : You can have soup with dinner.

q : You can have salad with dinner.

$p \oplus q$: You can have soup or salad, but not both.

In contrast, there is also the *exclusive or*: exactly one of the statements is true.

Definition

Let p and q be propositions. The **exclusive or** of p and q is the proposition “ p XOR q ”, denoted by “ $p \oplus q$ ”.

Example. Menu: Soup or Salad.

p : You can have soup with dinner.

q : You can have salad with dinner.

$p \oplus q$: You can have soup or salad, but not both.

Note

In mathematics, we don't use 'exclusive or' very often. Most of the time, **or** means **inclusive or**.

Truth Table

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	T	T	F
T	F	F	T	T
F	T	F	T	T
F	F	F	F	F

Truth Table

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	T		
T	F	F		
F	T	F		
F	F	F		

Question. What is

$p \wedge T$? It is (p and q) when q = T.

In case 1 and 3, p and (p and T) have the same truth values
so (p and T) = p.

Truth Table

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	T	T	
T	F	F	T	
F	T	F	T	
F	F	F	F	

Question. ~~What is~~ Simplify

$$p \wedge T? \quad p$$

$$p \vee T? \quad T$$

$$p \wedge F? \quad F$$

$$p \vee F? \quad p$$

Definition

Let p and q be propositions. The **conditional statement** $p \rightarrow q$ is the statement “If p , then q ”.

p implies q

If p happens, then q follows.

Definition

Let p and q be propositions. The **conditional statement** $p \rightarrow q$ is the statement “If p , then q ”.

Example.

- If Suki is sick, **then** she will stay at home.
- If it is sunny, **then** we go for a walk.
- If Linda passes the course, **then** $2 + 5 = 7$.
- If Linda passes the course, **then** $2 + 5 = 6$.

The Meaning of $p \rightarrow q$

- If p happens, then q must happen.
- If p does not happen, then q can happen or not.

The Meaning of $p \rightarrow q$

- If p happens, then q must happen.
- If p does not happen, then q can happen or not.

Example

If you don't eat, then you die.

p

q

$(p \rightarrow q)$ covers three cases:

p	q	
O	O	(you don't eat, and you die, 100% reasonable)
X (eat)	O	(you eat, and you die; it is still possible)
X (eat)	X	(you eat; and you don't die; it is also reasonable)

$(p \rightarrow q)$ does not cover:

O	X	(You don't eat, but you don't die; this is impossible)
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Truth Table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Remember: we want $(p \rightarrow q)$ to be T.

- If p is T, then in order for $(p \rightarrow q)$ to be T, q must be T.
- If p is F, then $(p \rightarrow q)$ is always T, no matter q is T or F.

There are a few other ways to express the conditional statement $p \rightarrow q$: **if p then q**

- p implies q .

- p only if q .

In contrast: $(p \text{ if } q)$ means $(\text{if } q \text{ then } p)$

Be careful: $(\text{if } q \text{ then } p)$ is different from $(\text{if } p \text{ then } q)$

There are a few other ways to express the conditional statement $p \rightarrow q$:

- p implies q .
- p only if q .
- q if p .
- q whenever p .
- q follows from p .
- p is sufficient for q .
- q is necessary for p .

In terms of logic, in $(p \rightarrow q)$:
 p is called the sufficient condition
 q is called the necessary condition

There are a few other ways to express the conditional statement $p \rightarrow q$:

- p implies q .
- p only if q .
- q if p .
- q whenever p .
- q follows from p .
- p is sufficient for q .
- q is necessary for p .
- $\neg p$ or q . (see next page)
- q unless $\neg p$.

Equality of logical statements

$$(p \rightarrow q) = (\neg p \vee q).$$

p	q	$p \rightarrow q$		$\neg p$	q	$\neg p \vee q$
T	T	T		F	T	T
T	F	F		F	F	F
F	T	T		T	T	T
F	F	T		T	F	T

same truth values

Definition

The **converse** of $p \rightarrow q$ is $q \rightarrow p$.

$(p \rightarrow q)$ and $(q \rightarrow p)$ are different statements:

Definition

The **converse** of $p \rightarrow q$ is $q \rightarrow p$.

The **inverse** of $p \rightarrow q$ is $\neg p \rightarrow \neg q$.

The **contra-positive** of $p \rightarrow q$ is $\neg q \rightarrow \neg p$.

Definition

The **converse** of $p \rightarrow q$ is $q \rightarrow p$.

The **inverse** of $p \rightarrow q$ is $\neg p \rightarrow \neg q$.

The **contra-positive** of $p \rightarrow q$ is $\neg q \rightarrow \neg p$.

Example. You die unless you eat.

if-then form: If you don't eat, then you die.

converse: If you die, then you don't eat. (This is meaningless)

inverse: If you eat, then you don't die. (It is possible, but it may not always happen)
(It can happen that: you eat, and you still die)

contra-positive:

If you don't die, then you eat. (This is the same as the original if-then statement)

Definition

If two compound propositions always have the same truth values, we say that they are **equivalent**.

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Exercise. $p \rightarrow q$ is equivalent to $\neg q \rightarrow \neg p$.

if-then

contra-positive

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

We write: $(p \rightarrow q) = (\neg q \rightarrow \neg p)$

$(q \rightarrow p) = (\neg p \rightarrow \neg q)$

converse

inverse

same truth values,
so they are equivalent (equal).

Definition

Let p and q be propositions. The **biconditional statement** $p \leftrightarrow q$ is the proposition " p if and only if q ".

(p if q) means (if q then p)
 $(q \rightarrow p)$

(p only if q) means (p implies q)
 $(p \rightarrow q)$

$(p \rightarrow q)$ and $(q \rightarrow p)$

Definition

Let p and q be propositions. The **biconditional statement** $p \leftrightarrow q$ is the proposition “ p if and only if q ”.

Example. Let

p : Maria learns Discrete Mathematics

q : Maria will find a good job

Express the statement $p \leftrightarrow q$ in good English.

Maria learns Discrete Mathematics if and only if she will find a good job

Example. Is the statement (the whole if-and-only-if statement)

$$x^2 = 4 \leftrightarrow x = 2$$

p

q

T or F?

The statement is $(p \leftrightarrow q) = (p \rightarrow q) \text{ and } (q \rightarrow p)$

$(q \rightarrow p) = (\text{if } x = 2, \text{ then } x^2 = 4)$ is T

$(p \rightarrow q) = (\text{if } x^2 = 4, \text{ then } x = 2)$ is F (because x can be -2)

So $(p \leftrightarrow q) = (p \rightarrow q) \text{ and } (q \rightarrow p) = F \text{ and } T = F$.

Example: $x^2 = 4 \leftrightarrow (x = 2 \text{ or } x = -2)$. Is this statement T or F ? T

p

q

r

$(p \leftrightarrow (q \text{ or } r)) = (p \rightarrow (q \text{ or } r)) \text{ and } ((q \text{ or } r) \rightarrow p)$ (Exercise)